

Expected bit rate of Golomb-Rice code for Geometric distribution

The bit rate $R_b(\delta)$ of a Golomb-Rice code with Rice parameter b for the codeword δ can be calculated as follows:

$$R_b(\delta) = b + 1 + \lfloor \frac{\delta}{2^b} \rfloor$$

The expectation value is: $E[R_b] = E[b + 1 + \lfloor \frac{\delta}{2^b} \rfloor]$

By linearity of the [expectation value](#):

$$E[R_b] = E[b + 1 + \lfloor \frac{\delta}{2^b} \rfloor] = E[b] + E[1] + E[\lfloor \frac{\delta}{2^b} \rfloor]$$

b and 1 are constants, so:

$$E[R_b] = b + 1 + E[\lfloor \frac{\delta}{2^b} \rfloor]$$

$E[\lfloor \frac{\delta}{2^b} \rfloor]$ can be calculated as follows:

$$E[\lfloor \frac{\delta}{2^b} \rfloor] = \sum_{j=0}^{\infty} \lfloor \frac{j}{2^b} \rfloor \cdot p(\delta = j)$$

assuming a [Geometric distribution](#) with parameter α for chance of failure and j failures until success for the probability, gives

$$= \sum_{j=0}^{\infty} \lfloor \frac{j}{2^b} \rfloor \cdot (1 - \alpha)\alpha^j$$

extracting constant terms out of the sum and getting rid of the floor function gives

$$= (1 - \alpha) \sum_{l=0}^{\infty} \sum_{j=l \cdot 2^b}^{(l+1) \cdot 2^b - 1} l \alpha^j = (1 - \alpha) \sum_{l=0}^{\infty} l \cdot \sum_{j=l \cdot 2^b}^{(l+1) \cdot 2^b - 1} \alpha^j$$

rearranging the second sum

$$= (1 - \alpha) \sum_{l=0}^{\infty} l \cdot \left(\sum_{j=0}^{(l+1) \cdot 2^b - 1} \alpha^j - \sum_{j=0}^{l \cdot 2^b - 1} \alpha^j \right)$$

with $\sum_{j=0}^n \alpha^j = \frac{1 - \alpha^{n+1}}{1 - \alpha}$ ([see](#)) and here $n = (l + 1) \cdot 2^b - 1$ or $n = l \cdot 2^b - 1$ we get

$$\begin{aligned} &= (1 - \alpha) \sum_{l=0}^{\infty} l \cdot \left(\frac{1 - \alpha^{(l+1)2^b}}{1 - \alpha} - \frac{1 - \alpha^{l \cdot 2^b}}{1 - \alpha} \right) \\ &= \sum_{l=0}^{\infty} l \cdot ((1 - \alpha^{(l+1)2^b}) - (1 - \alpha^{l \cdot 2^b})) \\ &= \sum_{l=0}^{\infty} l \cdot (1 - \alpha^{l \cdot 2^b} \alpha^{2^b} - 1 + \alpha^{l \cdot 2^b}) \end{aligned}$$

$$\begin{aligned}
&= \sum_{l=0}^{\infty} l \cdot (\alpha^{l \cdot 2^b} - \alpha^{l \cdot 2^b} \alpha^{2^b}) \\
&= \sum_{l=0}^{\infty} l \alpha^{l \cdot 2^b} \cdot (1 - \alpha^{2^b}) = (1 - \alpha^{2^b}) \cdot \sum_{l=0}^{\infty} l \alpha^{l \cdot 2^b} \\
&= (1 - \alpha^{2^b}) \cdot \sum_{l=0}^{\infty} l (\alpha^{2^b})^l
\end{aligned}$$

with $\sum_{l=0}^{\infty} l x^l = \frac{x}{(1-x)^2}$ (see) and here $x = \alpha^{2^b}$

$$\begin{aligned}
&= (1 - \alpha^{2^b}) \cdot \frac{\alpha^{2^b}}{(1 - \alpha^{2^b})^2} \\
&= \frac{\alpha^{2^b}}{1 - \alpha^{2^b}} = E(\lfloor \frac{\delta}{2^b} \rfloor)
\end{aligned}$$

with this result, the full expectation value is

$$E[R_b] = E[b + 1 + \lfloor \frac{\delta}{2^b} \rfloor] = b + 1 + E[\lfloor \frac{\delta}{2^b} \rfloor] = b + 1 + \frac{\alpha^{2^b}}{1 - \alpha^{2^b}}$$

expanding the 1 to simplify further

$$= b + \frac{1 - \alpha^{2^b}}{1 - \alpha^{2^b}} + \frac{\alpha^{2^b}}{1 - \alpha^{2^b}} = b + \frac{1 - \alpha^{2^b} + \alpha^{2^b}}{1 - \alpha^{2^b}}$$

and finally (under a Geometric distribution with probability of failure α)

$$E[R_b] = b + \frac{1}{1 - \alpha^{2^b}}$$

In our case we have $\alpha = 1 - \frac{1}{2^b}$ (because we have a fraction of $\frac{1}{2^b}$ nodes that can be "hit" by an edge) and therefore

$$E[R_b] = b + \frac{1}{1 - (1 - \frac{1}{2^b})^{2^b}}$$

and for large numbers 2^b , with the definition of the [exp-function](#), we finally get

$$E(R_b) \approx b + \frac{1}{1 - e^{-1}} \approx b + 1.58$$